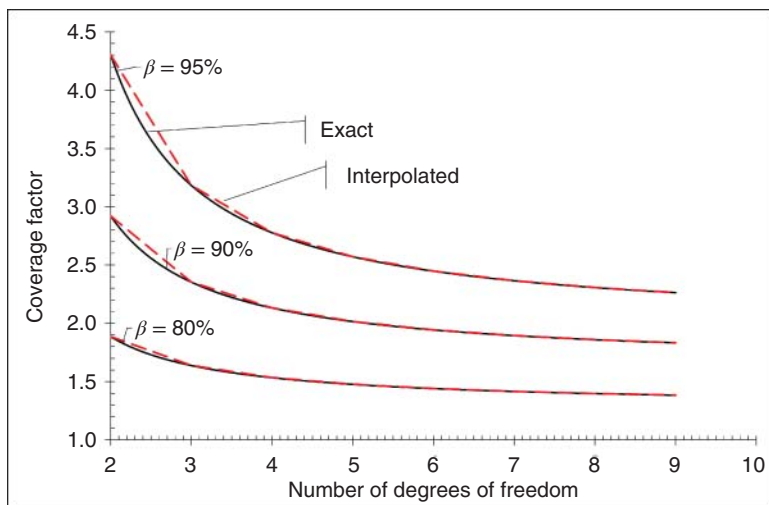




**Figure 5.15** Coverage factor as a function of the number of efficient measurements.

another or over time if series are scattered on different days. That is, the quantification procedure is not sensitive to slight variations between series. In the example of caffeine reported in Table 5.11, the  $A$  ratio varies from one validation material (level) to another, while it is more stable for theophylline. The analytical technique used, namely UHPLC-MS-MS, is known to be delicate to use, which probably explains these variations.

#### 5.4.4 Select Probability Values

Whether  $\beta$ -ETI or  $\beta$ - $\gamma$ -CTI is used, three values must be selected to construct and interpret the accuracy profile:

- $\beta$ : The coverage probability, i.e. the content proportion of future measurements of the tolerance interval, may be bound.
- $\gamma$ : The confidence level for the  $\beta$ - $\gamma$ -CTI, which represents the probability of a confidence interval. In statistical literature, the confidence level is usually denoted  $1 - \alpha$  where  $\alpha$  is the risk of error. When using  $\gamma$  instead, we are just conforming to the classic literature on statistical dispersion intervals.
- $[A_L; A_U]$ : The acceptance interval, also expressed as a percentage and defined by a probability of success.

The acceptance interval is centered on the nominal value of the validation material but is not always equal on both sides as some regulations may require asymmetric values, such as a recovery yield ranging between 80% and 110%, as detailed in Table 5.16. Regarding tolerance intervals, they are centered on the average inverse-predicted concentration.

This means they are generally symmetrical. But for some analytical methods, where the results are previously transformed into logarithms, the antilog-intervals

are no longer symmetrical. This situation is observable for polymerase chain reaction (PCR) methods applied to the detection of genetically modified organisms (GMO).

The choice of the three values listed above is application domain-specific and may not be the same for the methods applicable to the environment, consumer health, pharmaceutical industry, etc. For example, consider the values published by the FDA in the guidance on *Bioanalytical Method Validation* [6]. Two categories of analytical methods frequently employed in bioanalysis are considered: chromatographic assays (CC) and ligand binding assays (LBAs). The requirements of the section called “In study analysis” may help to define the three probability values in this case.

The FDA requirements are expressible in the compact form 4/6/15% for chromatographic assays and 4/6/20% for immuno-analyses. They are decoded as follows: four out of six quality controls (QC), or 67%, must be within  $\pm 15\%$  (or  $\pm 20\%$ ) acceptance interval around the nominal contents of the material used for the quality control. In other words, it is acceptable that a proportion of 67% of measurements made on a QC lie in the interval  $\pm 15\%$  (or  $\pm 20\%$ ) around the nominal value.  $\beta$ - $\gamma$ -CTI can also be used to establish some acceptance rules for the analysis quality control required by the US-FDA in the context of biological analysis.

#### Summary of Table 1 of FDA Guidance *Bioanalytical Method Validation*.

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##### *Quality Controls (QC)*

###### Elements:

- $\geq 4$  QC levels (LLOQ, low, medium, and high) and  $\geq 2$  replicates per QC level in each analytical run.
- Total QCs should be 5% of unknown samples or  $\geq 6$ , whichever number is greater.
- If the analytical runs consist of distinct processing batches, the QC acceptance criteria should be applied for the whole run and each distinct batch within the runs.

###### Acceptance criteria:

- CC:  $\geq 67\%$  (4/6) of QCs should be  $\pm 15\%$  of the nominal, and  $\geq 50\%$  of QCs per level should be  $\pm 15\%$  of their nominal.
- LBA:  $\geq 67\%$  (4/6) of QCs should be  $\pm 20\%$  of the nominal, and  $\geq 50\%$  of QCs per level should be  $\pm 20\%$  of their nominal.

##### **Trueness<sup>a)</sup> and precision**

###### Trueness: between runs:

- CC:  $\pm 15\%$  of nominal concentrations
- LBA:  $\pm 20\%$  of nominal concentrations

###### Precision: between runs:

- CC:  $\pm 15\%$  CV
  - LBA:  $\pm 20\%$  CV
- 

a) Beware: in the original document the term “accuracy” is used instead of “trueness,” conversely to the VIM definitions.

Source: Adapted from FDA [6].

**Table 5.12** THEOPHYLLINE – half tolerance intervals for different probability values.

| Parameters                       | Concentration ( $\mu\text{g/l}$ ) |       |       |       |       |       |
|----------------------------------|-----------------------------------|-------|-------|-------|-------|-------|
|                                  | 0.05                              | 0.1   | 0.5   | 1     | 2.5   | 10    |
| Acceptance probability           | 20%                               | 20%   | 20%   | 20%   | 20%   | 20%   |
| ITE- $\beta$ (80%)               | 33.1%                             | 17.9% | 9.9%  | 12.5% | 15.0% | 7.0%  |
| ITC- $\beta$ - $\gamma$ (80–95%) | 58.6%                             | 30.6% | 17.6% | 22.5% | 24.2% | 12.1% |
| ITC- $\beta$ - $\gamma$ (67–95%) | 44.6%                             | 23.2% | 13.3% | 17.1% | 18.4% | 9.2%  |

The names of the intervals are abbreviated as in the text and the associated probabilities appear in brackets.

Official FDA requirement is that at least 4 out of 6, i.e. 67% of measurements made on QC samples, fall within the acceptance interval  $[A_L; A_U]$ , which is, depending on the type of method, either  $\pm 15\%$  or  $\pm 20\%$  of the reference (or nominal) value.

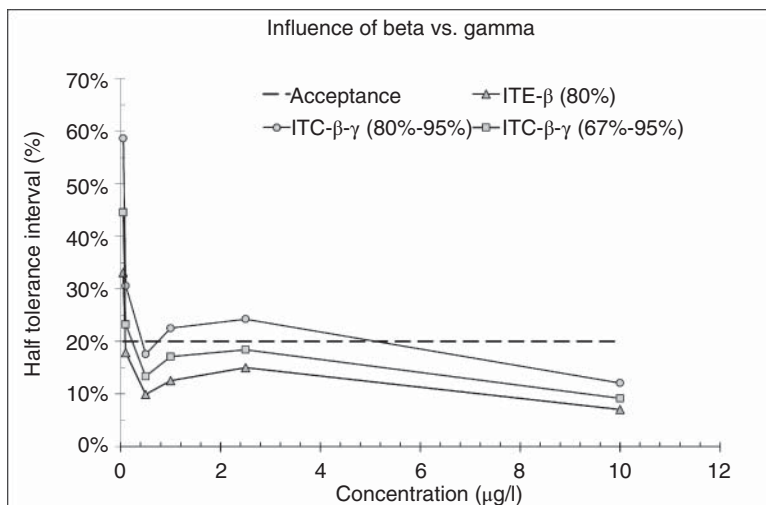
To have a margin of safety and to be sure of achieving this objective, it may be recommended to choose a probability of 80% per level so that at least 4 out of 5 or 80% (instead of more than 4 out of 6) measurements are included in the chosen acceptance interval. Finally, according to FDA rules, the probability values associated with the statistical dispersion intervals, which are used to verify whether a validated method is suitable for QC, come down to the following:

- Content proportion  $\beta$ ‰: 80% or 67%.
- Confidence level  $\gamma$ ‰: 95%.
- Acceptance:  $\pm 15\%$  or  $\pm 20\%$

Table 5.12 combines the half-intervals, expressed in % and calculated from the THEOPHYLLINE dataset, for various combinations of probability and interval types. To restore the coverage of one of the intervals in measurement units, simply apply the percentage to the concentration in the top row. For example, the half  $\beta$ -ETI (80%) is 33.1% for a concentration of 0.05  $\mu\text{g/l}$ , which gives the following bonds for the coverage interval for 80% of future measurements:

$$[0.05 \times (1 - 0.331); 0.05 \times (1 + 0.331)] = [0.033; 0.067]$$

Figure 5.16 shows this set of values in graphical form, which shows the respective roles of the coverage probability  $\beta$  and the confidence level  $\gamma$ . If the three concentration levels of the QCs are correctly chosen, then this method of determining theophylline perfectly satisfies the FDA requirements for quality control. Indeed, between 0.13 and 10  $\mu\text{g/l}$ , the half intervals are below the acceptance limit symbolized by the dashed line at  $\pm 20\%$ . From this example, it is also clear that changing  $\beta$ ‰ from 67% to 80% has a significative influence on the width of the interval at all concentrations.



**Figure 5.16** THEOPHYLLINE – half tolerance intervals for different probability values. Probabilities are in brackets after interval name abbreviations.

#### 5.4.5 Select the Type of Tolerance Interval

Referring to the literature, both types of intervals- $\beta$ -ETI and  $\beta$ - $\gamma$ -CTI-have been proposed to conduct method validation. The following table summarizes the main arguments and literature now attached to either type of interval. In the last row appears the type of TI recommended in this book.

|              | $\beta$ -Expectation interval ( $\beta$ -ETI)                                 | $\beta$ - $\gamma$ -Content interval ( $\beta$ - $\gamma$ -CTI)                 |
|--------------|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Objective    | Predict the interval that contains a known proportion of future observations. | Define the limits of a specified proportion of a distribution with a guarantee. |
| References   | [17–20]                                                                       | [27]                                                                            |
| Applications | Method validation and/or estimation of MU                                     | Control charts and/or routine quality control                                   |

As previously mentioned, the  $\beta$ -ETI is thus particularly suited to predicting an interval computed from a set of observations that contains a given proportion of future measurements. It therefore, predicts whether the method can produce, for a given sample, a high proportion of acceptable measurements, i.e. within the limits of a predefined acceptance interval. In practice, a recommended value of  $\beta\% = 80\%$  was adopted because it allows compliance with the requirement formulated by the FDA, as explained in the previous chapter. In the Section 7.3, which deals with control charts, it is also shown that this proportion allows us to follow the typical requirements encountered for standardized control charts.

### 5.4.6 Proportion of Nonacceptable Measures

A classic misunderstanding about the  $\beta$  proportion is to believe that only  $\beta\%$  acceptable measurements values are produced by the method. This is an incorrect interpretation of this probability. The correct interpretation is that, at least,  $\beta\%$  results are within the tolerance interval if the  $\beta$ -ETI remains within the acceptance interval bounds. While this condition is true, much more than  $\beta\%$  produced measurement values are acceptable.

To illustrate this point, it is possible to predict the proportion of acceptable measurement values that will be inside the acceptance interval  $[A_L, A_U]$ . Consider the parameters listed in Table 5.3 page 118 for the THEOPHYLLINE study. The second validation material contains 0.5  $\mu\text{g/l}$ , and the acceptance interval is  $\pm 25\%$ , giving:

|                                    |                                                                          |
|------------------------------------|--------------------------------------------------------------------------|
| Inverse-predicted grand mean       | $\bar{\bar{Z}} = 0.520$                                                  |
| Limits of the acceptance interval  | $[A_L, A_U] = [0.5 - 25\%; 0.5 + 25\%]$<br>$[A_L, A_U] = [0.375, 0.625]$ |
| Standard deviation of $\beta$ -ETI | $s_{IT} = 0.0350$                                                        |
| Number of degrees of freedom       | $N_E = 7.02$                                                             |

The method consists in calculating the differences between the inverse-predicted grand mean and the two limits of the acceptance interval. If they are divided by the standard deviation of the  $\beta$ -ETI, these two quantities denoted  $t_{inf}$  and  $t_{sup}$  can be considered as approximate quantiles of a Student's law with  $N_E$  degrees of freedom. In this example,  $N_E = 7.02$ , and allows to obtain the probability associated with each quantile.

| Quantile of the Student's $t$ law                                                 | Associated probability |
|-----------------------------------------------------------------------------------|------------------------|
| $t_{sup} = \frac{ \bar{\bar{Z}} - A_U }{s_{IT}} = \frac{0.1458}{0.0350} = 4.1292$ | 0.22%                  |
| $t_{inf} = \frac{ A_L - \bar{\bar{Z}} }{s_{IT}} = \frac{0.1054}{0.0350} = 3.0106$ | 0.98%                  |

When the theophylline concentration is around 0.5  $\mu\text{g/l}$ , the global risk of producing non-acceptable measures is  $0.98\% + 0.22\% = 1.2\%$  which is much below the traditional value of 5% used for statistical testing. It also means that, at this concentration, the method can produce about 98.8% of acceptable measures.

Table 5.13 generalizes this calculation for the different concentration levels. The global risk of *nonacceptable* results very rapidly decreases while concentration increases. The breakpoint of more than 25% of nonacceptable measures lies between 0.05 and 0.1  $\mu\text{g/l}$  where the LOQ is located. To achieve this calculation, Student's law table that provides the probability associated with noninteger number of degrees of freedom is necessary.

As explained for Resource H worksheet it is possible to obtain an approximate value by interpolation using the built-in function **TDIST** that gives this probability. **TINV** is the inverse function of **TDIST**. To obtain the percentage of non-acceptable